MTH603 Assignment # 02 \solution Fall 2023/2024

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Question 1 **10 Marks**

Form the table given below

x	1	2	3	4	5
$f(x)$	1.000	1.4142	1.7320	2.000	2.2361

Estimate the value f (1.5) using appropriate Newton's formula.

Solution:

Newton's forward difference interpolation method to find solution

Newton's forward difference table is

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1	**1**				
		0.4142			
2	1.4142		**-0.0964**		
		0.3178		**0.0466**	
3	1.732		-0.0498		**-0.0287**
		0.268		0.0179	
4	2		-0.0319		
		0.2361			
5	2.2361				

Newton's forward difference interpolation formula is

$$y(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\cdot\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\cdot\Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!}\cdot\Delta^4 y_0$$

$$p = \frac{x - x_0}{h} = \frac{x-1}{1} = x - 1$$

Substituting these values, we get

$$y(x) = 1 + (x-1)\times 0.4142 + \frac{(x-1)(x-2)}{2}\times -0.0964 + \frac{(x-1)(x-2)(x-3)}{6}\times 0.0466 + \frac{(x-1)(x-2)(x-3)(x-4)}{24}\times -0.0287$$

$$y(x) = 1 + 0.4142\times(x-1) + (-0.0482)\times\left(x^2 - 3x + 2\right) + 0.0078\times\left(x^3 - 6x^2 + 11x - 6\right) + (-0.0012)\times\left(x^4 - 10x^3 + 35x^2 - 50x + 24\right)$$

$$y(x) = 1 + (0.4142x - 0.4142) + \left(- 0.0482x^2 + 0.1446x - 0.0964 \right) + \left(0.0078x^3 - 0.0466x^2 + 0.0854x - 0.0466 \right) + \left(- 0.0012x^4 + 0.012x^3 - 0.0419x^2 + 0.0598x - 0.0287 \right)$$

$$y(x) = - 0.0012x^4 + 0.0197x^3 - 0.1367x^2 + 0.704x + 0.4141$$

Now, substitute $x = 1.5$

$$y(1.5) = - 0.0012 \times 1.5^4 + 0.0197 \times 1.5^3 - 0.1367 \times 1.5^2 + 0.704 \times 1.5 + 0.4141 = 1.2232$$

Question 2 — 10 Marks

If the size of a bacteria colony at different time is given by following table, estimate the size of colony at 0.015

Time	0.01	0.02	0.03	0.04	0.05
Size	98.4342	48.4392	31.7775	23.4492	18.4542

Solution:

Solution:
The value of table for x and y

x	0.01	0.02	0.03	0.04	0.05
y	98.4342	48.4392	31.7775	23.4492	18.4542

Newton's forward difference interpolation method to find solution

Newton's forward difference table is

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0.01	**98.4342**				
		-49.995			
0.02	48.4392		**33.3333**		
		-16.6617		**-24.9999**	
0.03	31.7775		8.3334		**19.9998**
		-8.3283		-5.0001	
0.04	23.4492		3.3333		
		-4.995			
0.05	18.4542				

Newton's forward difference interpolation formula is

$$y(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!} \cdot \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \cdot \Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!} \cdot \Delta^4 y_0$$

$$p = \frac{x - x_0}{h} = \frac{x - 0.01}{0.01} = 100x - 1$$

$$p = \frac{x - x_0}{h} = \frac{x - 0.01}{0.01} = 100x - 1$$

Substituting these values, we get

$$y(x) = 98.4342 + (100x - 1) \times \ - 49.995 + \frac{(100x - 1)(100x - 2)}{2} \times 33.3333 + \frac{(100x - 1)(100x - 2)(100x - 3)}{6} \times \ - 24.9999 + \frac{(100x - 1)(100x - 2)(100x - 3)(100x - 4)}{24} \times 19.9998$$

$$y(x) = 98.4342 + (- 49.995) \times (100x - 1) + 16.6667 \times \left(10000x^2 - 300x + 2\right) + (- 4.1667) \times \left(1000000x^3 - 60000x^2 + 1100x - 6\right) + 0.8333 \times \left(100000000x^4 - 10000000x^3 + 350000x^2 - 5000x + 24\right)$$

$$y(x) = 98.4342 + (- 4999.5x + 49.995) + \left(166666.5x^2 - 4999.995x + 33.3333\right) + \left(- 4166650x^3 + 249999x^2 - 4583.315x + 24.9999\right) + \left(83332500x^4 - 8333250x^3 + 291663.75x^2 - 4166.625x + 19.9998\right)$$

$$y(x) = 83332500x^4 - 12499900x^3 + 708329.25x^2 - 18749.435x + 226.7622$$

Now, substitute $x = 0.015$

$$y(0.015) = 83332500 \times 0.015^4 - 12499900 \times 0.015^3 + 708329.25 \times 0.015^2 - 18749.435 \times 0.015 + 226.7622 = 66.9263$$